Technical Documentation

ACDT Project 1, GROUP32: Is daily social media use associated with student mental health?
Interactive data story (scroll, filters, calculators): dhkim060225-web.github.io/group32-sns-mental-health
Code: github.com/dhkim060225-web/group32-sns-mental-health

1. Research question and hypotheses

Outcome Y = Mental_Health_Score; treatment X = Avg_Daily_Usage_Hours. Model: $Y = \beta_0 + \beta_1 X + \varepsilon$. H_0: $\beta_1 = 0$; H_1: $\beta_1 \neq 0$. Our earlier version was one-sided ($\beta_1 < 0$, controlling for academic level). We switched to two-sided because the regression framework tests $\beta_1 \neq 0$, a two-sided test is the more cautious choice, and with reverse causation possible we did not want to rule out either direction in advance. **$\alpha = 0.05$**: a Type I error would justify blanket screen-time rules and wasted counselling time; a Type II error would leave struggling heavy users unflagged. Neither cost dominates, so we kept the conventional level.

2. Data and preprocessing

Source: *Students' Social Media Addiction* (Kaggle), 705 rows, 13 self-reported variables, no missing values; dataset description mirrored at baselight.app/u/kaggle/dataset/rehan497_students_social_media_addiction. Recruitment method, sampling frame and consent process are not documented (treated as a convenience sample); the file is de-identified. Student IDs 242 and 274 are identical in every answer; one was removed (**n = 704**). 110 countries: 79 (72%) contribute a single student, and the 31 countries with two or more students hold 625 students. For DML, Country was recoded to the 10 largest countries plus "Other".

Variable	Type / range	Role	Reason
Avg_Daily_Usage_Hours	Continuous, 1.5–8.5 h	Treatment (X)	The conventional wisdom is about time spent
Mental_Health_Score	Integer, observed 4–9 (1–10 scale)	Outcome (Y)	Replaces GPA, which is not in the data
Age, Gender, Academic_Level, Relationship_Status	Background	Control	Fixed before usage; can confound, not caused by usage
Country	110 categories	Control / cluster	Strongest confounder; breaks independence
Most_Used_Platform	12 categories	Moderator	Imperfect proxy for purpose of use, not a direct measure
Sleep_Hours_Per_Night	Continuous	Mediator	Usage can reduce sleep; not controlled for the total effect
Addicted_Score, Conflicts_Over_Social_Media	Scores / counts	Not controlled	Likely consequences of usage (bad controls)
Affects_Academic_Performance	Yes / No	Secondary outcome	Closest proxy for the original GPA question

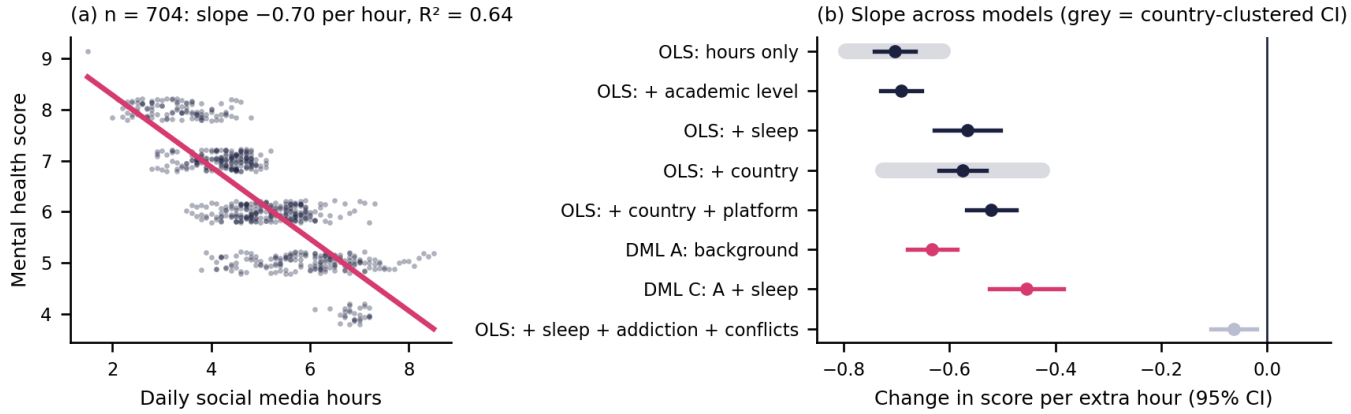
3. Methods

Framework level	Method	Implementation details
Sampling distribution / Inference	OLS, two-sided t-test, 95% CI	df = 702, t* = 1.963. Country breaks independence, so CR1 cluster-robust SEs (110 clusters) are reported for the pooled slope and for the within-country (country fixed effects) slope. Shapiro–Wilk on residuals; Spearman rank correlation as an ordinal check.
Prediction	Held-out validation	200 random 80/20 splits (seeds 0–199) and 200 repeats of 5-fold splits that hold out whole countries; test R^2 and RMSE vs a train-mean baseline. Prediction intervals: $\hat{y} \pm t^* \cdot s\sqrt{(1 + 1/n + (x-\bar{x})^2/S_{xx})}$.
Inference (heterogeneity)	Per-platform OLS; nested F-test	Bootstrap CIs (2,000 resamples). F-test: platform-specific vs common slope with platform intercepts (n ≥ 10, varying outcome). Bonferroni across the 9 platform slope tests (α = 0.0056).
Reasoning	Fixed effects; mediation; sensitivity	110 country dummies. Sleep mediation by product of coefficients a·b, 1,000 bootstrap resamples. Robustness value for an unmeasured confounder (Cinelli & Hazlett, 2020) from the simple-regression t-statistic.
Reasoning	Double/debiased ML (partially linear model)	$Y = \theta X + g(W) + \varepsilon$, $X = m(W) + v$ (Chernozhukov et al., 2018). Random forests (100 trees, min leaf 5; 8 in subgroups), 5-fold cross-fitting, 3 repetitions, median aggregation; influence-function SE that assumes independent students.

4. Results

Model	Estimate per hour	95% CI	R^2
OLS: hours only	−0.704	−0.74 to −0.66 (clustered: −0.80 to −0.61)	0.642
OLS: + academic level	−0.691	−0.73 to −0.65	0.657
OLS: + sleep (direct effect)	−0.566	−0.63 to −0.50	0.657
OLS: + country fixed effects (within-country)	−0.575	−0.62 to −0.53 (clustered: −0.73 to −0.42)	0.848

Model	Estimate per hour	95% CI	R^2
OLS: + country + platform	−0.521	−0.57 to −0.47	0.874
OLS: + sleep + addiction + conflicts	−0.063	−0.11 to −0.02	0.896
DML A: background (age, gender, level, relationship, country)	−0.633	−0.68 to −0.59	–
DML B: A + platform	−0.589	−0.65 to −0.53	–
DML C: A + sleep	−0.454	−0.52 to −0.38	–
DML D: A + addiction + conflicts	0.056	0.01 to 0.10	–



Validation: random splits give mean test R^2 0.635 (range 0.484–0.730), RMSE 0.662 vs baseline 1.104; holding out whole countries gives 0.623 (0.588–0.638), RMSE 0.678. Adjusted R^2: hours 0.641, country 0.629, both 0.819. Platform heterogeneity: $F(8, 650) = 3.59$, $p = 0.0004$; all 9 platform slopes $p \leq 0.0014$ (Bonferroni passed); academic level $p = 0.81$; gender $p = 0.31$. Mediation: a = −0.709, b = 0.194, indirect = −0.138 (95% CI −0.177 to −0.104), 19.6% of total. Spearman ρ = −0.802 (Pearson −0.801). Sensitivity: robustness value 0.71 to erase the slope, 0.48 to halve it. DML subgroups: Undergraduate −0.63, Graduate −0.68, High School −0.70, Female −0.63, Male −0.69, Instagram −0.66, TikTok −0.54, Facebook −0.66; WhatsApp could not be estimated because scores are constant within its countries. Secondary outcome: students reporting academic harm use 5.54 vs 3.80 h (Welch $t = 25.4$, $p < 0.001$).

Model D: controlling for addiction and conflicts removes the effect because both are probably downstream of usage and strongly tied to the outcome (r = −0.95 for addiction). We report it only to show why such variables must not be controlled.

5. Limitations and threats to validity

- **External validity:** undocumented, non-random sample; 72% of countries contribute one student, so the within-country estimate rests on 31 countries. Held-out validation stays inside this dataset.
- **Internal validity:** cross-sectional self-reports; reverse causation and unmeasured confounding (stress, workload, purpose of use) remain. DML adjusts only for measured covariates, its intervals ignore clustering, and no valid instrument was available.
- **Measurement:** platform is an imperfect proxy for purpose of use, so platform differences do not show that passive and active use differ.
- **Data quality:** identical scores for all users within each of LINE, KakaoTalk and VKontakte and within 13% of countries with n ≥ 2; academic-impact answers move from 0% to 100% across usage bands; correlations far stronger than in large real-world studies (Orben & Przybylski, 2019).
- **Ethics:** public, de-identified data with an undocumented consent process; results should not inform decisions about individual students.

6. Reproducibility

Repository: **analysis.py** (all computations, writes results.json), **results.json**, **data/**, **index.html** (website), **tools/** (page and document builders). Python 3 with pandas, numpy, scipy, scikit-learn. Run **python analysis.py**, then **python tools/build_page.py** and the two document builders. All random procedures use fixed seeds; re-running analysis.py from a clean copy reproduced results.json exactly.

References

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. The Econometrics Journal, 21(1), C1–C68.

Cinelli, C., & Hazlett, C. (2020). Making sense of sensitivity: Extending omitted variable bias. Journal of the Royal Statistical Society: Series B, 82(1), 39–67.

Hale, L., & Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: A systematic literature review. Sleep Medicine Reviews, 21, 50–58.

Orben, A., & Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. Nature Human Behaviour, 3(2), 173–182.